

Kunci jawaban Assignment #1 : Teori Produksi

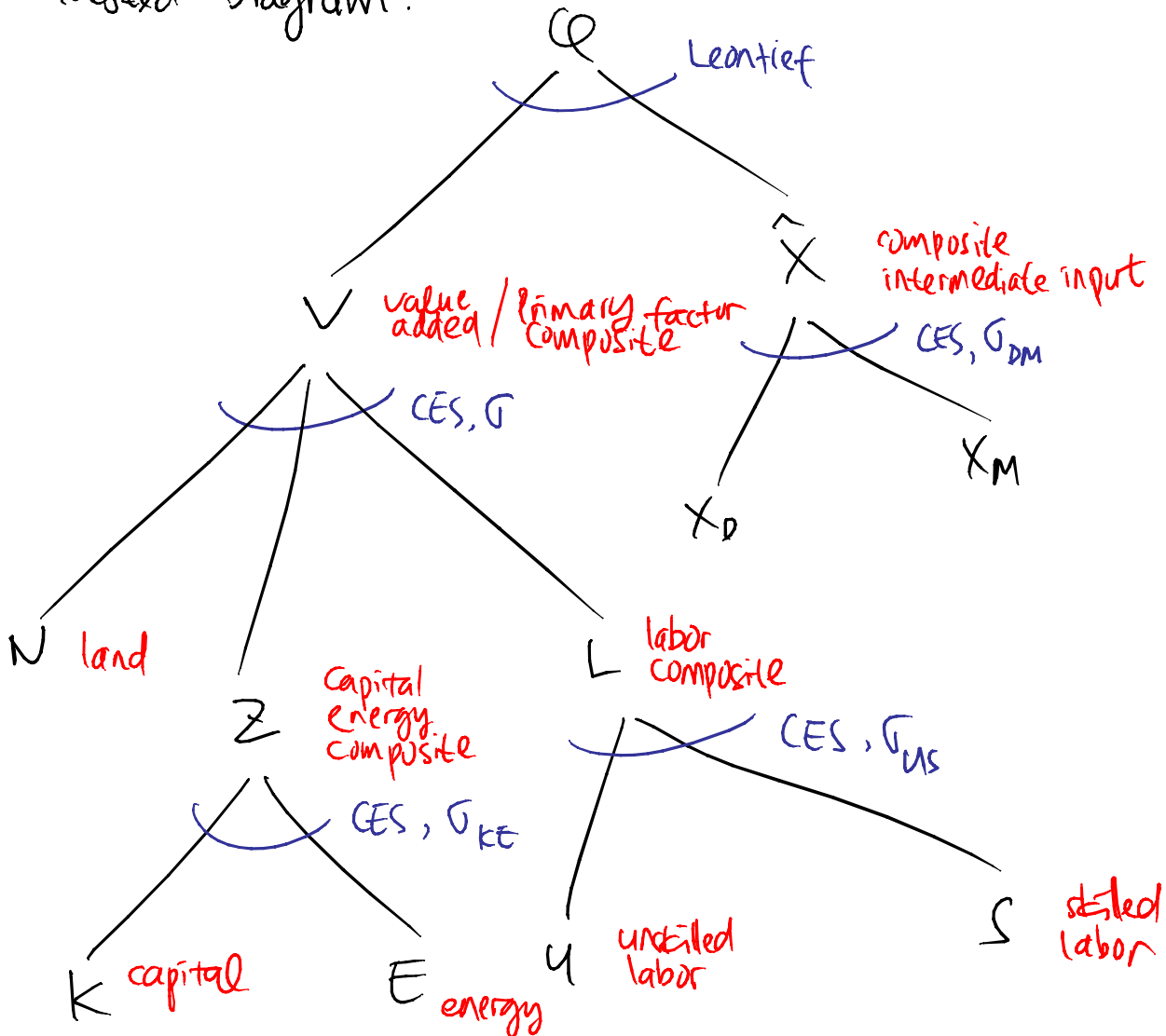
Note Title

6/9/2013

Minimize cost: $C = w_U U + w_S S + p_N N + rK + p_E E + p_D X_D + p_M X_M$

Subject to: $Q = \min \left\{ \frac{f_{\sigma} (f_{\sigma_{US}} (U, S), f_{\sigma_{KE}} (K, E), N)}{a_1}, \frac{f_{\sigma_{DM}} (X_D, X_M)}{a_2} \right\}$

Nested Diagram:



DEMAND FOR LABOR BY SKILL TYPE

① $U = L \delta_u^{\sigma_{us}} \left(\frac{w_u}{w} \right)^{-\sigma_{us}}$

② $S = L \delta_s^{\sigma_{us}} \left(\frac{w_s}{w} \right)^{\sigma_{us}}$

③ $w = \left(\delta_u^{\sigma_{us}} \frac{1-\sigma_{us}}{w_u} + \delta_s^{\sigma_{us}} \frac{1-\sigma_{us}}{w_s} \right)^{\frac{1}{1-\sigma_{us}}}$

DEMAND FOR CAPITAL AND ENERGY

$$(4) \quad K = z \delta_K^{\sigma_{KE}} \left(\frac{r}{p_z} \right)^{-\sigma_{KE}}$$

$$(5) \quad E = z \delta_E^{\sigma_{KE}} \left(\frac{p_E}{p_z} \right)^{-\sigma_{KE}}$$

$$(6) \quad p_z = \left(\delta_K^{\sigma_{KE}} r^{1-\sigma_{KE}} + \delta_E^{\sigma_{KE}} p_E^{1-\sigma_{KE}} \right)^{\frac{1}{1-\sigma_{KE}}}$$

DEMAND FOR LAND, ϕ , AND L

$$(7) \quad N = V \delta_N^{\sigma} \left(\frac{p_N}{p_v} \right)^{-\sigma} \quad (8) \quad z = V \delta_z^{\sigma} \left(\frac{p_z}{p_v} \right)^{-\sigma} \quad (9) \quad L = V \delta_L^{\sigma} \left(\frac{w}{p_v} \right)^{-\sigma}$$

$$(10) \quad p_v = \left(\delta_N^{\sigma} p_N^{1-\sigma} + \delta_z^{\sigma} p_z^{1-\sigma} + \delta_L^{\sigma} w^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$$

DEMAND FOR DOMESTIC AND IMPORTED INTERMEDIATE

$$(11) \quad X_D = \hat{X} \delta_D^{\sigma_{DM}} \left(\frac{p_D}{\hat{p}_X} \right)^{-\sigma_{DM}}$$

$$(12) \quad X_M = \hat{X} \delta_M^{\sigma_{DM}} \left(\frac{p_M}{\hat{p}_X} \right)^{-\sigma_{DM}}$$

$$(13) \quad \hat{p}_X = \left(\delta_D^{\sigma_{DM}} p_D^{1-\sigma_{DM}} + \delta_M^{\sigma_{DM}} p_M^{1-\sigma_{DM}} \right)^{\frac{1}{1-\sigma_{DM}}}$$

DEMAND FOR PRIMARY FACTOR AND COMPOSITE INTERMEDIATE

$$(14) \quad V = a_1 Q$$

$$(15) \quad \hat{X} = a_2 Q$$